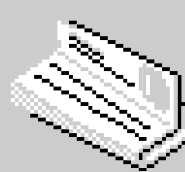
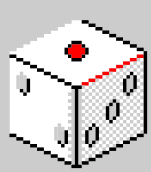
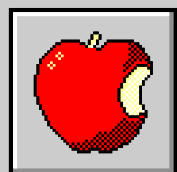


Perception

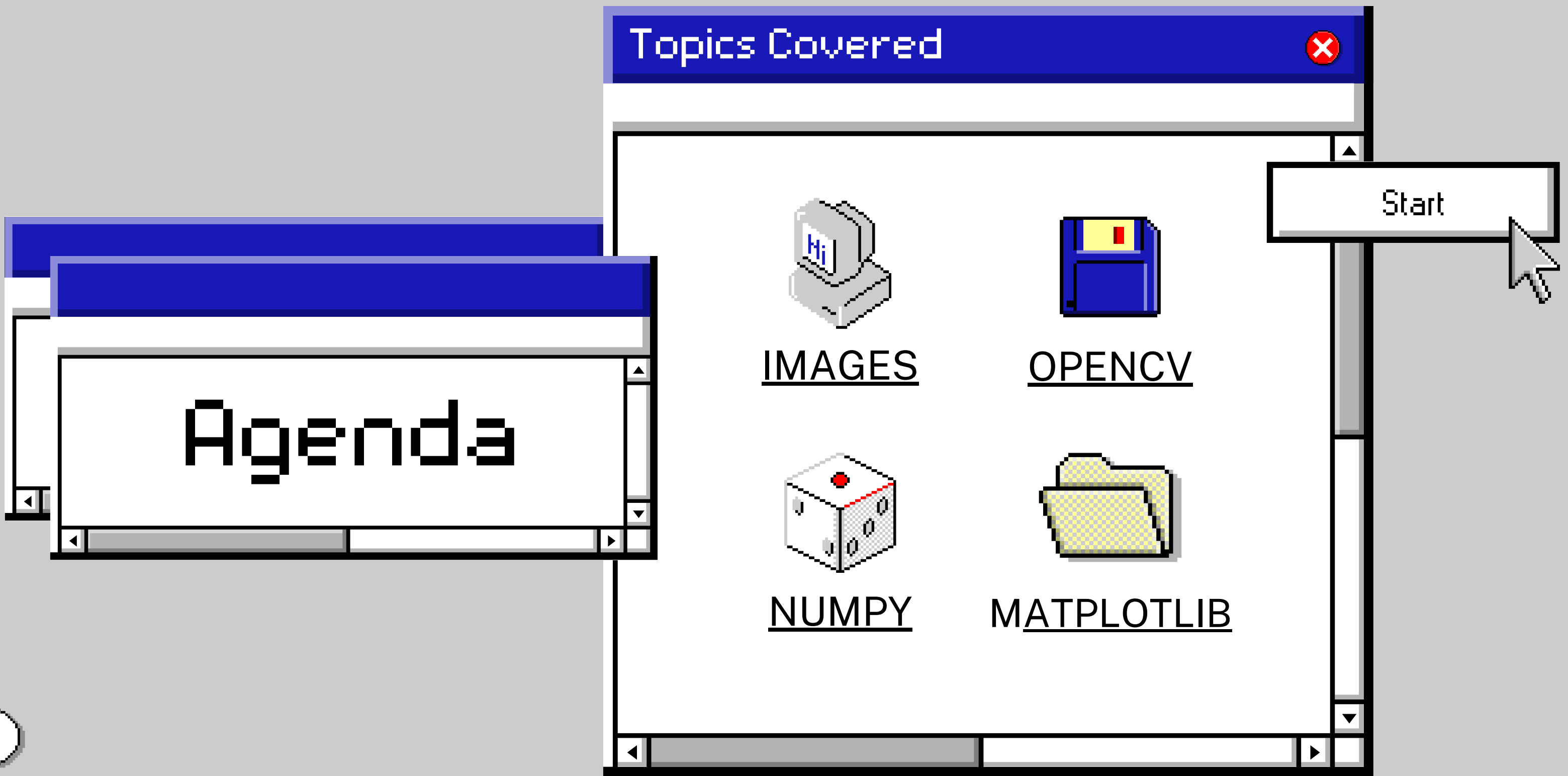
DAY - 1



AYUSH BHUSHAN



11:11PM

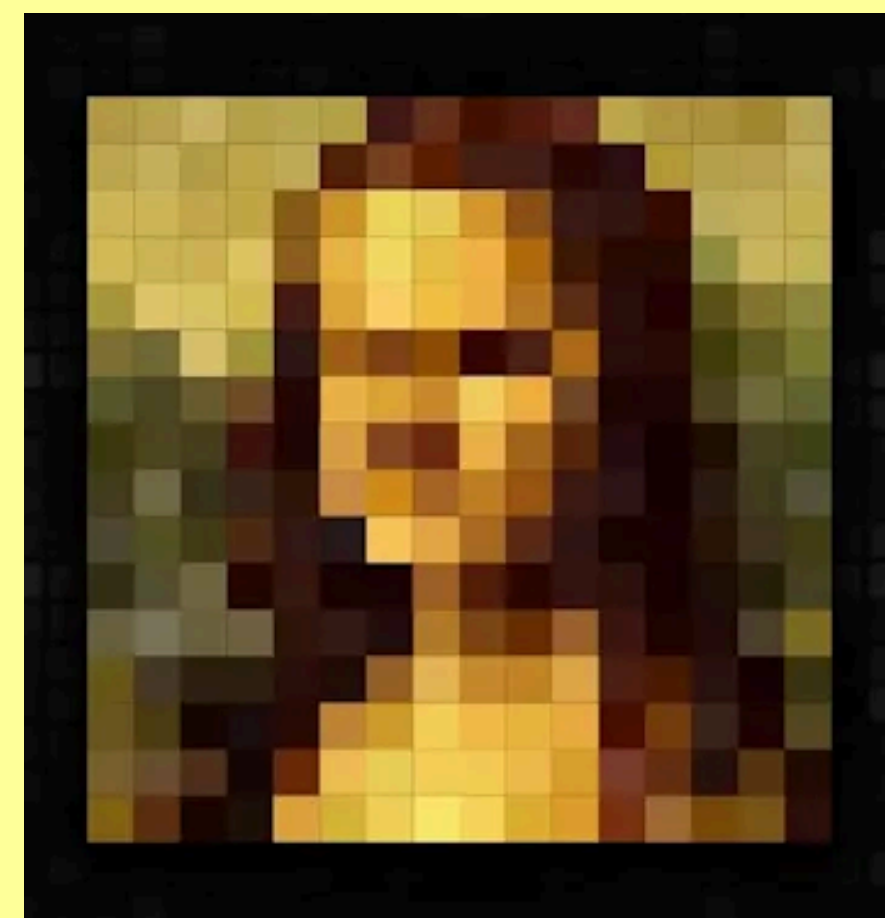




Image

Pixels

- ## Numbers



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Image Creation

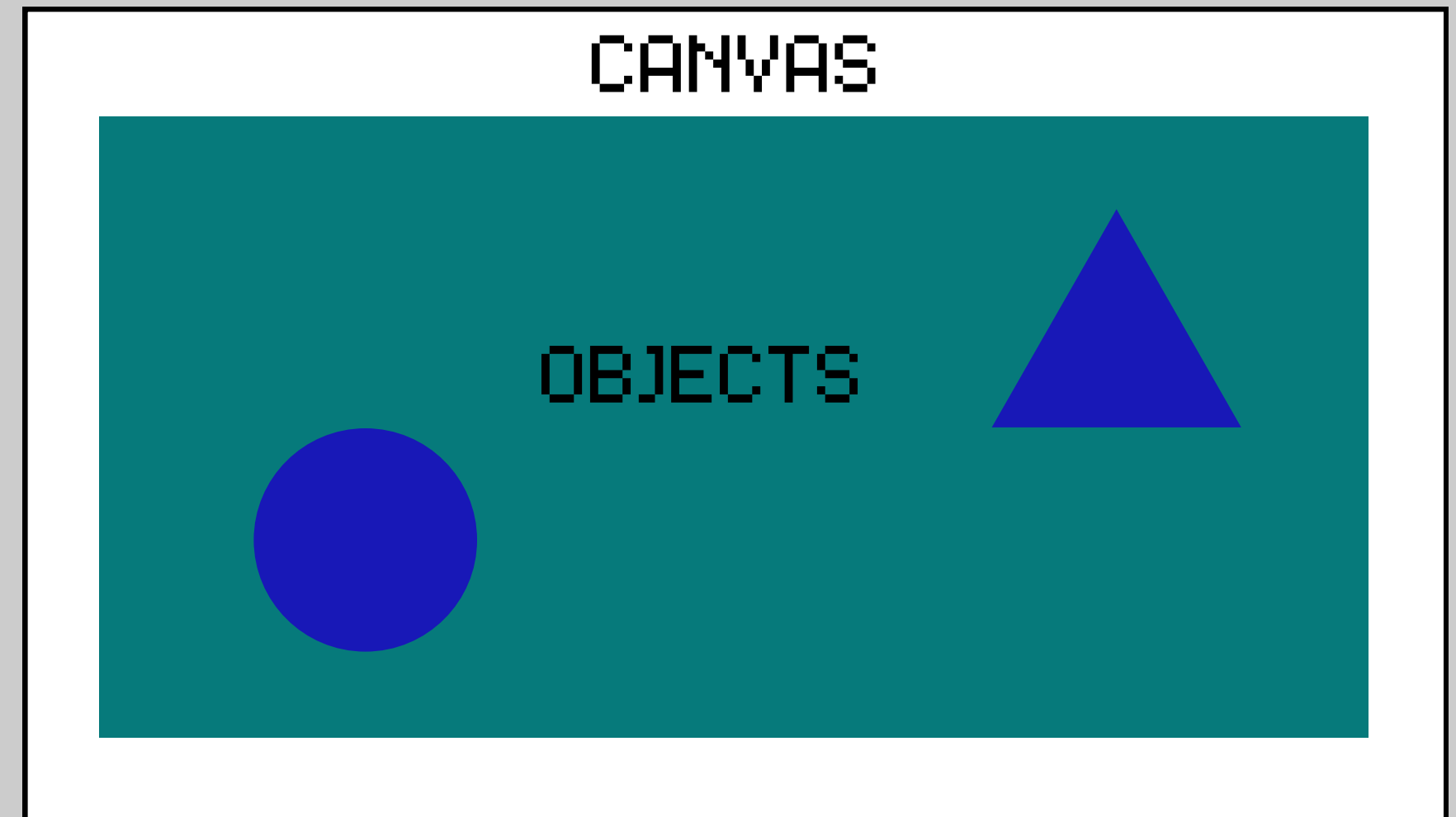
```
np.full(shape, fill_value, dtype)
```

shape - (height, width, channels)

fill_value - (B,G,R) **

dtype= np.uint8 ⚠

```
import cv2
import numpy as np
img=np.full((500,500,3),(0,0,255),dtype=np.uint8)
cv2.imshow("Image1",img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

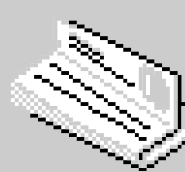
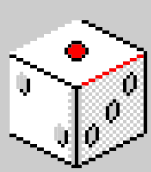
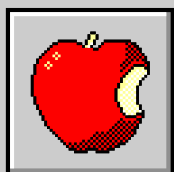
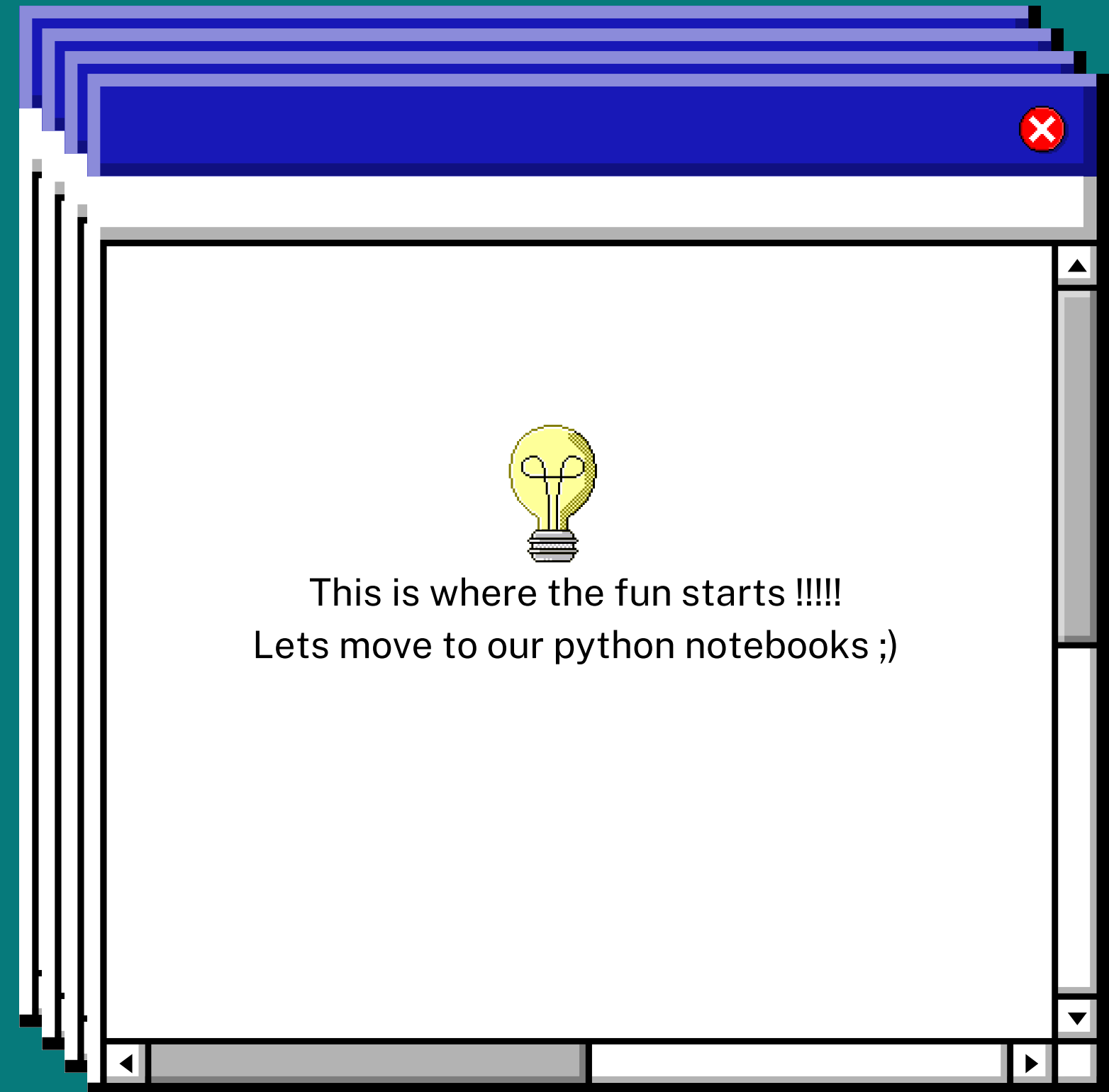


- imshow - displays the image in a separate window
- waitKey - it waits for the set time before moving forward (0 means forever)
- destroyAllWindows - closes all display windows

Whiteboard Page



Drawing different shapes



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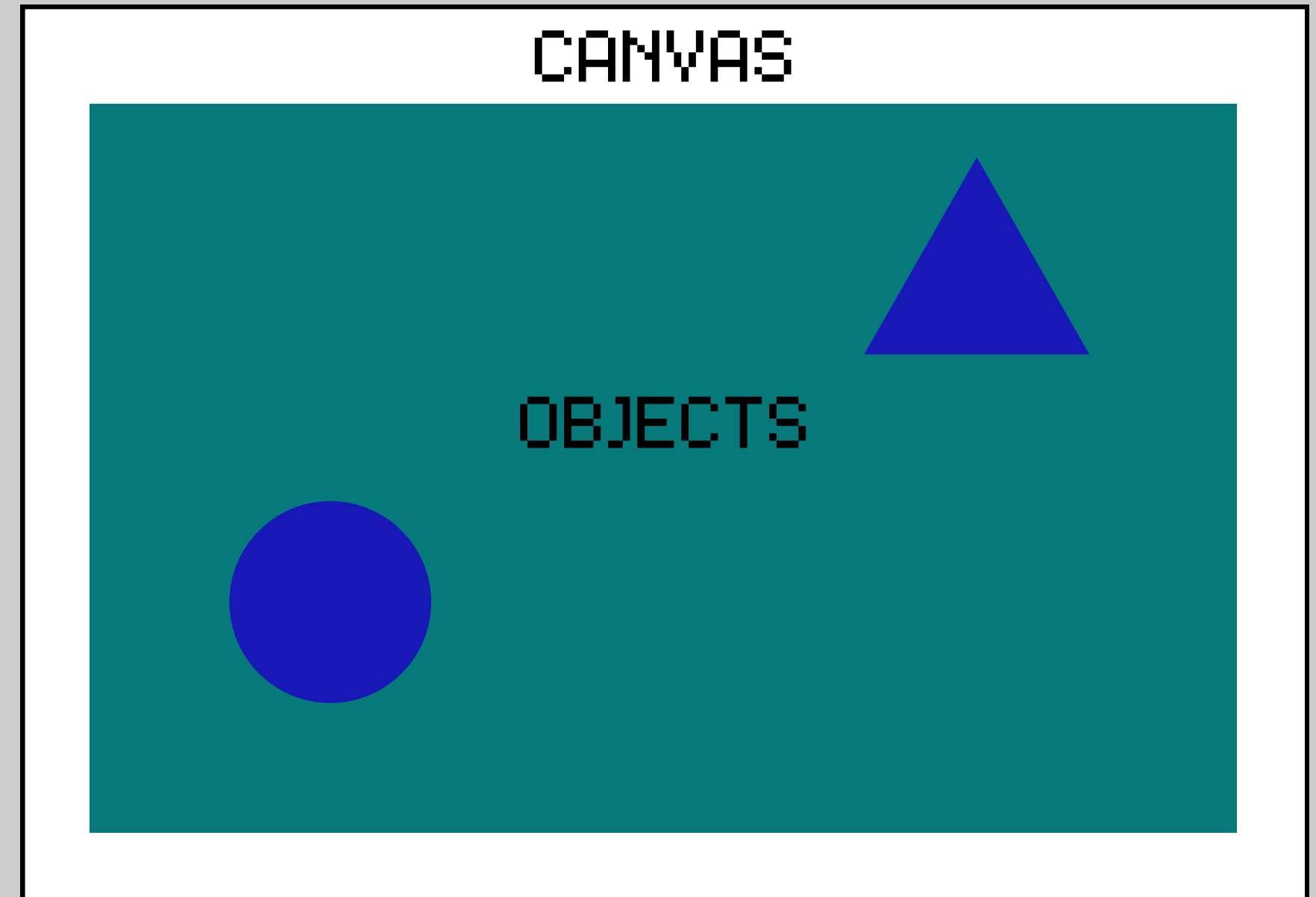
Image Processing

Loading an Image on our computer

```
cv2.imread(filename(file directory),flag)
```

flag parameters -

- cv2.IMREAD_COLOR - loads a color image
- cv2.IMREAD_GRAYSCALE - loads a b/w image



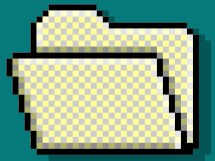
Resize

Rotate

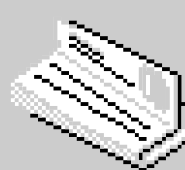
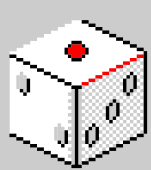
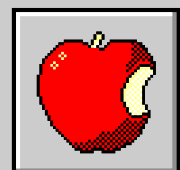
Padding

Masking

THRESHOLDING



I will use this slide to explain - its supposed to be empty only <3



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VIDEOS

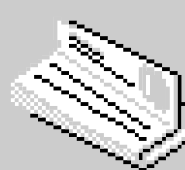
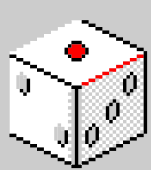
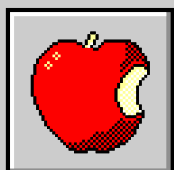


In simple words, Videos are a sequence of many images playing at a specific frame rate

Frame Rate is the number of images captured per second. Our normal phone cameras have usually 24, 30, 60 fps.



Lets move straight to a simple and cool demonstration



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Thank you!

I hope I didn't bore y'all :)

